S1. Evidence map and locked data roles

The study has two different evidential roles. CREMA-D's ten original test actors have already been inspected. All new CREMA-D results are exploratory. RAVDESS is an external reserve whose selection, model family, thresholds and evaluation script were locally hashed before feature inference. No RAVDESS examples fit experts, calibration maps, support limits or tuning thresholds.

The CREMA router/calibration/tune roles contain nine actors each. Twenty-four eligible clips per actor are chosen by SHA-256 of a fixed prefix plus clip ID. For these 648 clips, clean, central 25% occlusion, and blur 2 inputs produce 1,944 condition-specific observations. Conditions from the same clip and actor remain dependent; they are not independent calibration sources.

The RAVDESS subset uses all 24 actors and six shared intended-emotion classes. Four full-AV speech clips per actor/class are chosen by a separate fixed filename hash, giving 576 clips. Emotion codes 01, 03, 04, 05, 06, 07 map to neutral, happy, sad, angry, fearful and disgust. Calm and surprise are excluded by the six-class head vocabulary before evaluation. Neither outcome quality nor media appearance participates in selection.

Integrity and access. The official Zenodo record and source archive checksums are saved. Downloaded ZIP members are checked with their central-directory CRC32; local SHA-256 values identify the exact media used. Whole-archive MD5 values are provenance metadata, not claimed as verified. The transfer optimization is recorded in a dated pre-inference deviation. The first feature-inference log points to the fitted artifact lock. The locks are local audit evidence, not third-party registration.

Independence limits. Actor grouping prevents actor reuse between local roles. It does not prove absence of unknown pretrained-encoder exposure or semantic duplication across Internet corpora. Both datasets measure intended acted expressions. Neither provides physical humanoid execution, natural source occlusion, or a newly collected within-CREMA confirmation set.

S2. Observable representation and exact models

The expert order is audio, interaction fusion, video. Each six-class logit vector is centered and divided by its own standard deviation, floored at10^{-8}, then converted to softmax probabilities. Features are three maxima, three top-two margins, three normalized entropies, three pairwise Jensen–Shannon divergences, and three top-one agreements. Pair order is audio/fusion, audio/video, fusion/video.

The eight added visual measurements are the mean and maximum cosine distance between adjacent frame embeddings; mean and maximum cosine distance between an original frame and its mild crop; central luminance standard deviation; central mean absolute spatial gradient; central mean absolute temporal difference; and central mean luminance. All use decoded 224-square letterboxed RGB frames in[0,1]. Eight frames are uniformly spaced. Four alternate frames are center-cropped at pixels 11:213 and bilinearly resized to 224 square. The central image statistics use pixels 56:168 in each spatial dimension, and luminance is the simple RGB mean.

Frozen WavLM Base yields a 768-dimensional temporal mean; DINOv2 ViT-B/14 yields a 768-dimensional average over frames. Existing expert heads use two 768-to-96 projections, layer normalization and GELU. Interaction fusion uses two modality tokens and four attention heads; the classifier maps the concatenated 192 dimensions through 96 hidden units to 6 classes. The same preprocessing and expert-head checkpoints are retained under visual transformations. Seeds 11 and 22 are both evaluated; they share pretrained encoders and data roles.

All new histogram boosting models have 100 iterations, 7 leaf nodes, minimum leaf size 20, learning rate .05, L2=1, no early stopping and seed 56. Actor-uniform weights are constructed on the router role before disagreement restriction and renormalized for each fitting subset. Direct regression estimates signed gain. The three-class classifier estimates$\Delta \in \{-1, 0, 1\}$. Conditional heads estimate non-neutrality and rescue conditional on non-neutrality. No model uses actor identity, corruption labels, target labels, or clean-reference features at inference.

S3. Three policies, calibration, and certificate

For each method with signed score$s$, the positive batch policy considers only disagreement clips with$s > 0$. It sorts by descending score, breaks ties lexicographically by clip ID, and retains the longest actor-weighted prefix within budget. It does not refill the remaining budget with predicted negative or zero gains. For unequal indivisible weights this greedy prefix is not asserted to solve an arbitrary knapsack optimum.

The independent threshold is the smallest nonnegative score level whose strict exceedance mass on clean tuning actors is at most the nominal budget. Whole ties are excluded together. The threshold is frozen under visual shift. This constrains tuning-set change volume, not future population volume.

Support-aware routing standardizes the eight raw reliability features using clean router data, with standard deviations floored at10^{-5}. Squared standardized distance is compared with the 95th percentile of clean calibration distances. Outside support the score is set to-1 and audio is retained. This is a heuristic rejection rule, not an exchangeability or uncertainty guarantee.

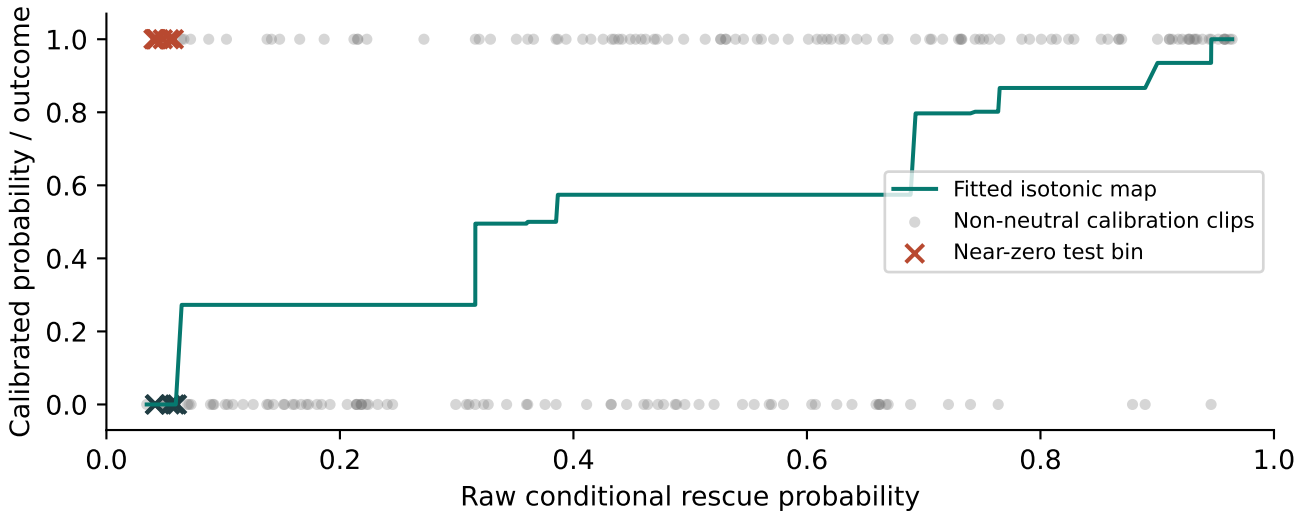
Isotonic maps fit each conditional probability on the calibration role. Sigmoid alternatives fit logistic regression with C=1 to the clipped probability logit; clipping is$[10^{-6}, 1 - 10^{-6}]$. Direct outcome probabilities are calibrated classwise and renormalized before computing rescue minus harm. All three conditions participate in calibration for these new maps; the support reference and operating thresholds use clean clips only.

For the separate certificate, split positive direct-gain values at their median on clean router observations, intersect with visual support, and fix the two regions before calibration outcomes. For region$j$, source contributionY_{gj} is mean signed gain including zeros outside the region. With independent sources,$Y_{gj} \in [-1, 1]$. Simultaneous Hoeffding lower endpoints subtract$\sqrt{2\log(2/.05)/9}$. Both endpoints are negative, so the certified policy retains audio everywhere. The assumptions concern the clean calibration population and do not certify RAVDESS or arbitrary corrupted inputs.

S4. Near-zero rescue predictions on inspected CREMA-D

These 13 cases were diagnosed after the original test was inspected. Eight are rescues. Twelve receive exactly zero conditional rescue probability from an isotonic plateau; none lies outside that map's fitted input range. These labels were not used to refit the new calibration maps. Target and expert predictions are integer class indices in ANG,DIS,FEA,HAP,NEU,SAD order.

Clip	Gain	Raw r	Mapped r	Rescue p
1001_ITS_DIS_XX	1	0.0431	0.0000	0.0000
1001_IWL_DIS_XX	1	0.0483	0.0000	0.0000
1020_TAI_NEU_XX	-1	0.0605	0.0574	0.0545
1036_IWL_FEA_XX	-1	0.0585	0.0000	0.0000
1036_IWW_ANG_XX	1	0.0539	0.0000	0.0000
1058_ITS_NEU_XX	-1	0.0415	0.0000	0.0000
1058_IWL_NEU_XX	-1	0.0544	0.0000	0.0000
1058_MTI_SAD_XX	1	0.0425	0.0000	0.0000
1058_TAI_SAD_XX	1	0.0413	0.0000	0.0000
1058_TIE_FEA_XX	1	0.0416	0.0000	0.0000
1060_IEO_ANG_MD	-1	0.0420	0.0000	0.0000
1060_TIE_SAD_XX	1	0.0403	0.0000	0.0000
1085_TIE_ANG_XX	1	0.0577	0.0000	0.0000

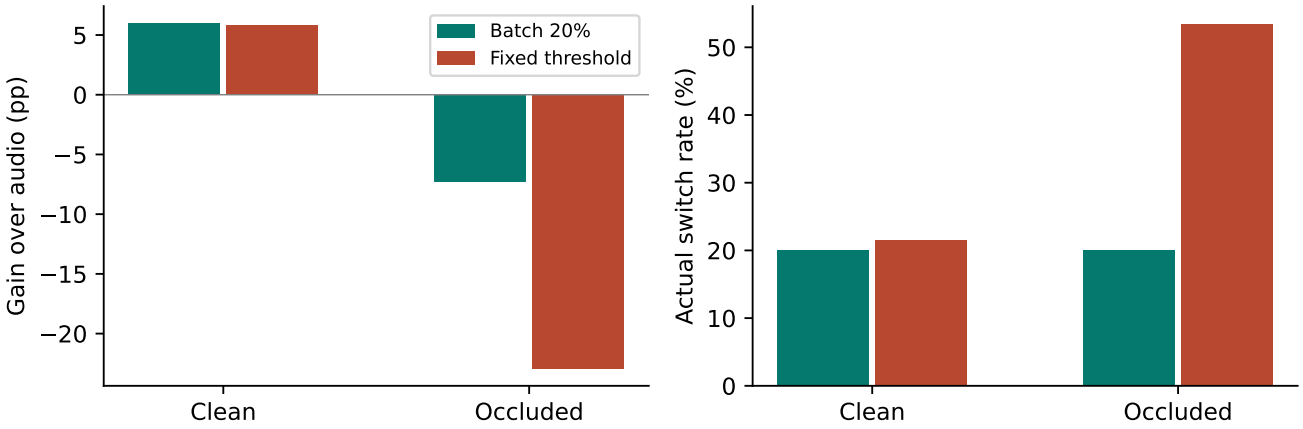


The zero plateau is supported by six non-neutral calibration clips from four actors, all harms. Its empirical conditional outcome is not a population certainty. This support audit distinguishes an internal fitted plateau from out-of-range clipping, and explains why a reliability plot alone is insufficient to locate the failure.

S5. Switch volume versus selected quality

Let v be change mass and u mean signed gain among changes. The symmetric decomposition$\Delta G = \overline{u}\Delta v + \overline{v}\Delta u$ is exact arithmetic; it is not a causal mediation estimate. The inspected CREMA policy moves from 21.46%to 53.41%changes and from +5.85 to -22.93 points. The volume term is -2.50 points and the selected-quality term is -26.28 points. The corrected positive-budget rule changes none of the selected examples at the original 20%operating point.

Transition	N	R0	H0	R1	H1	N1
retained_switch	138	67	34	41	34	63
new_switch	300	21	35	15	210	75
abandoned_switch	38	21	6	7	4	27
retained_audio	344	24	29	5	97	242



R0/H0 count rescues/harms before the visual intervention;R1/H1 count them after. N1 is neutral correctness change after intervention. A neutral prediction replacement can have both experts wrong. The mean selector score increases by 0.119, with an exploratory paired actor interval about[0.018,0.223], while correction outcomes worsen.

S6. Exploratory CREMA-D: seed 11, clean

All entries are percentages or accuracy points. The original test is inspected, so this table is diagnostic rather than fresh confirmation. Both modes use the nominal 20%budget; fixed thresholds can change a different fraction on this sample. Each policy remains fitted without these test outcomes.

positive_batch

Method	Acc.	Gain	Change	R	H	Neutral
af_gain	62.07	2.32	9.76	4.51	2.20	3.05
scores_gain	62.32	2.56	10.00	5.00	2.44	2.56
reliable_gain	64.39	4.63	17.68	8.78	4.15	4.76
outcome_raw	64.02	4.27	17.07	8.41	4.15	4.51
outcome_isotonic	63.41	3.66	10.12	5.37	1.71	3.05
two_raw	64.88	5.12	18.54	9.27	4.15	5.12
two_isotonic	64.51	4.76	17.07	8.54	3.78	4.76
two_platt	64.15	4.39	16.22	8.05	3.66	4.51
supported_gain	64.27	4.51	17.44	8.66	4.15	4.63
certified_gain	59.76	0.00	0.00	0.00	0.00	0.00
margin	66.59	6.83	20.00	10.85	4.02	5.12
confidence	66.22	6.46	20.00	10.61	4.15	5.24
entropy	65.98	6.22	20.00	10.61	4.39	5.00

fixed

Method	Acc.	Gain	Change	R	H	Neutral
af_gain	62.07	2.32	9.76	4.51	2.20	3.05
scores_gain	62.32	2.56	10.00	5.00	2.44	2.56
reliable_gain	64.39	4.63	17.68	8.78	4.15	4.76
outcome_raw	64.02	4.27	17.07	8.41	4.15	4.51
outcome_isotonic	63.41	3.66	10.12	5.37	1.71	3.05
two_raw	64.88	5.12	18.54	9.27	4.15	5.12
two_isotonic	64.51	4.76	17.07	8.54	3.78	4.76
two_platt	64.15	4.39	16.22	8.05	3.66	4.51
supported_gain	64.27	4.51	17.44	8.66	4.15	4.63
certified_gain	59.76	0.00	0.00	0.00	0.00	0.00
margin	66.34	6.59	16.95	9.88	3.29	3.78
confidence	66.10	6.34	18.29	10.00	3.66	4.63
entropy	65.98	6.22	18.90	10.12	3.90	4.88

The support rejection fraction is observable from the feature distribution, but it is not a direct estimate of semantic visibility. The audio fallback establishes the zero-gain, zero-switch-harm reference. A rule that abstains everywhere cannot be credited with recovering hidden content.

S7. Exploratory CREMA-D: seed 11, video_occlude50

All entries are percentages or accuracy points. The original test is inspected, so this table is diagnostic rather than fresh confirmation. Both modes use the nominal 20%budget; fixed thresholds can change a different fraction on this sample. Each policy remains fitted without these test outcomes.

positive_batch

Method	Acc.	Gain	Change	R	H	Neutral
af_gain	55.24	-4.51	20.00	3.66	8.17	8.17
scores_gain	54.88	-4.88	20.00	3.29	8.17	8.54
reliable_gain	57.07	-2.68	14.02	2.32	5.00	6.71
outcome_raw	57.07	-2.68	17.44	3.54	6.22	7.68
outcome_isotonic	59.63	-0.12	2.56	0.73	0.85	0.98
two_raw	55.00	-4.76	20.00	3.17	7.93	8.90
two_isotonic	56.46	-3.29	16.95	3.17	6.46	7.32
two_platt	57.44	-2.32	14.02	2.68	5.00	6.34
supported_gain	59.76	0.00	0.00	0.00	0.00	0.00
certified_gain	59.76	0.00	0.00	0.00	0.00	0.00
margin	56.59	-3.17	20.00	4.39	7.56	8.05
confidence	56.71	-3.05	20.00	4.27	7.32	8.41
entropy	55.49	-4.27	20.00	3.90	8.17	7.93

fixed

Method	Acc.	Gain	Change	R	H	Neutral
af_gain	55.00	-4.76	21.22	3.90	8.66	8.66
scores_gain	54.39	-5.37	21.22	3.66	9.02	8.54
reliable_gain	57.07	-2.68	14.02	2.32	5.00	6.71
outcome_raw	57.07	-2.68	17.44	3.54	6.22	7.68
outcome_isotonic	59.63	-0.12	2.56	0.73	0.85	0.98
two_raw	54.27	-5.49	21.59	3.41	8.90	9.27
two_isotonic	56.46	-3.29	16.95	3.17	6.46	7.32
two_platt	57.44	-2.32	14.02	2.68	5.00	6.34
supported_gain	59.76	0.00	0.00	0.00	0.00	0.00
certified_gain	59.76	0.00	0.00	0.00	0.00	0.00
margin	48.05	-11.71	36.71	5.98	17.68	13.05
confidence	48.78	-10.98	37.07	6.22	17.20	13.66
entropy	49.39	-10.37	37.32	6.59	16.95	13.78

The support rejection fraction is observable from the feature distribution, but it is not a direct estimate of semantic visibility. The audio fallback establishes the zero-gain, zero-switch-harm reference. A rule that abstains everywhere cannot be credited with recovering hidden content.

S8. Exploratory CREMA-D: seed 22, clean

All entries are percentages or accuracy points. The original test is inspected, so this table is diagnostic rather than fresh confirmation. Both modes use the nominal 20%budget; fixed thresholds can change a different fraction on this sample. Each policy remains fitted without these test outcomes.

positive_batch

Method	Acc.	Gain	Change	R	H	Neutral
af_gain	62.07	1.22	10.49	4.51	3.29	2.68
scores_gain	63.17	2.32	15.12	6.83	4.51	3.78
reliable_gain	63.78	2.93	19.27	9.02	6.10	4.15
outcome_raw	63.90	3.05	20.00	9.15	6.10	4.76
outcome_isotonic	63.78	2.93	14.88	7.07	4.15	3.66
two_raw	64.27	3.41	20.00	9.39	5.98	4.63
two_isotonic	64.02	3.17	18.54	8.66	5.49	4.39
two_platt	64.51	3.66	20.00	9.51	5.85	4.63
supported_gain	64.02	3.17	19.02	9.02	5.85	4.15
certified_gain	60.85	0.00	0.00	0.00	0.00	0.00
margin	64.51	3.66	20.00	9.76	6.10	4.15
confidence	65.24	4.39	20.00	10.12	5.73	4.15
entropy	65.37	4.51	20.00	10.24	5.73	4.02

fixed

Method	Acc.	Gain	Change	R	H	Neutral
af_gain	62.07	1.22	10.49	4.51	3.29	2.68
scores_gain	63.17	2.32	15.12	6.83	4.51	3.78
reliable_gain	63.78	2.93	19.27	9.02	6.10	4.15
outcome_raw	63.54	2.68	21.59	9.63	6.95	5.00
outcome_isotonic	63.78	2.93	14.88	7.07	4.15	3.66
two_raw	64.39	3.54	21.83	10.24	6.71	4.88
two_isotonic	64.02	3.17	18.54	8.66	5.49	4.39
two_platt	64.39	3.54	21.83	10.24	6.71	4.88
supported_gain	64.02	3.17	19.02	9.02	5.85	4.15
certified_gain	60.85	0.00	0.00	0.00	0.00	0.00
margin	64.39	3.54	20.12	9.76	6.22	4.15
confidence	65.24	4.39	20.00	10.12	5.73	4.15
entropy	65.61	4.76	20.85	10.73	5.98	4.15

The support rejection fraction is observable from the feature distribution, but it is not a direct estimate of semantic visibility. The audio fallback establishes the zero-gain, zero-switch-harm reference. A rule that abstains everywhere cannot be credited with recovering hidden content.

S9. Exploratory CREMA-D: seed 22, video_occlude50

All entries are percentages or accuracy points. The original test is inspected, so this table is diagnostic rather than fresh confirmation. Both modes use the nominal 20%budget; fixed thresholds can change a different fraction on this sample. Each policy remains fitted without these test outcomes.

positive_batch

Method	Acc.	Gain	Change	R	H	Neutral
af_gain	57.07	-3.78	17.20	3.29	7.07	6.83
scores_gain	58.78	-2.07	9.39	1.83	3.90	3.66
reliable_gain	56.83	-4.02	18.78	3.90	7.93	6.95
outcome_raw	56.46	-4.39	16.83	3.29	7.68	5.85
outcome_isotonic	59.02	-1.83	7.44	1.46	3.29	2.68
two_raw	55.98	-4.88	16.59	2.80	7.68	6.10
two_isotonic	57.93	-2.93	12.20	2.20	5.12	4.88
two_platt	56.59	-4.27	15.12	2.68	6.95	5.49
supported_gain	60.85	0.00	0.00	0.00	0.00	0.00
certified_gain	60.85	0.00	0.00	0.00	0.00	0.00
margin	57.56	-3.29	20.00	4.39	7.68	7.93
confidence	57.93	-2.93	20.00	4.51	7.44	8.05
entropy	57.07	-3.78	20.00	4.02	7.80	8.17

fixed

Method	Acc.	Gain	Change	R	H	Neutral
af_gain	57.07	-3.78	17.20	3.29	7.07	6.83
scores_gain	58.78	-2.07	9.39	1.83	3.90	3.66
reliable_gain	56.83	-4.02	18.78	3.90	7.93	6.95
outcome_raw	56.46	-4.39	16.83	3.29	7.68	5.85
outcome_isotonic	59.02	-1.83	7.44	1.46	3.29	2.68
two_raw	55.98	-4.88	16.59	2.80	7.68	6.10
two_isotonic	57.93	-2.93	12.20	2.20	5.12	4.88
two_platt	56.59	-4.27	15.12	2.68	6.95	5.49
supported_gain	60.85	0.00	0.00	0.00	0.00	0.00
certified_gain	60.85	0.00	0.00	0.00	0.00	0.00
margin	45.73	-15.12	44.02	6.59	21.71	15.73
confidence	45.61	-15.24	44.15	6.83	22.07	15.24
entropy	45.73	-15.12	44.02	6.83	21.95	15.24

The support rejection fraction is observable from the feature distribution, but it is not a direct estimate of semantic visibility. The audio fallback establishes the zero-gain, zero-switch-harm reference. A rule that abstains everywhere cannot be credited with recovering hidden content.

S10. External RAVDESS: seed 11, clean

The subset contains 576 clips from 24 actors, with 24 clips per actor. No outcome-based exclusions are made. The policies and clean tuning thresholds were frozen before external feature inference. All entries below are percentages or accuracy points.

Method	Acc.	Gain	Change	R	H	Neutral
af_gain	37.33	2.95	9.38	3.47	0.52	5.38
scores_gain	37.50	3.12	9.03	3.47	0.35	5.21
reliable_gain	34.72	0.35	2.26	0.35	0.00	1.91
outcome_raw	35.24	0.87	3.82	1.04	0.17	2.60
outcome_isotonic	34.38	0.00	0.35	0.00	0.00	0.35
two_raw	36.11	1.74	5.73	1.91	0.17	3.65
two_isotonic	35.24	0.87	3.82	1.04	0.17	2.60
two_platt	35.24	0.87	3.12	0.87	0.00	2.26
supported_gain	34.38	0.00	0.00	0.00	0.00	0.00
certified_gain	34.38	0.00	0.00	0.00	0.00	0.00
margin	38.54	4.17	14.06	5.56	1.39	7.12
confidence	39.06	4.69	16.84	6.42	1.74	8.68
entropy	38.19	3.82	19.44	6.77	2.95	9.72
legacy_opportunity_advantage	39.24	4.86	20.31	7.81	2.95	9.55
legacy_logistic	38.37	3.99	18.40	6.60	2.60	9.20
legacy_gb_gain	38.89	4.51	15.97	6.42	1.91	7.64
legacy_rf_gain	37.33	2.95	11.11	4.34	1.39	5.38
legacy_mlp	38.37	3.99	17.71	6.94	2.95	7.81

Positive at-most batch, same nominal budget

Method	Gain	CI low	CI high	Change
af_gain	2.95	0.87	5.03	9.38
scores_gain	3.12	1.22	5.03	9.03
reliable_gain	0.35	0.00	0.87	2.26
outcome_raw	0.87	-0.17	2.26	3.82
outcome_isotonic	0.00	0.00	0.00	0.35
two_raw	1.74	0.69	2.95	5.73
two_isotonic	0.87	0.00	1.91	3.82
two_platt	0.87	0.17	1.74	3.12
supported_gain	0.00	0.00	0.00	0.00
certified_gain	0.00	0.00	0.00	0.00
margin	4.69	1.74	7.29	19.27
confidence	4.86	2.08	7.29	19.97
entropy	3.82	0.87	6.60	19.97

Fixed-threshold intervals resample paired actors with the policy fixed. Batch intervals recompute the rank-constrained selection within each resampled batch. They are conditional on frozen experts and router fitting. The two modes do not have identical deployment semantics. Full budget sweeps and raw endpoints are in the CSV files.

S11. External RAVDESS: seed 11, video_occlude50

The subset contains 576 clips from 24 actors, with 24 clips per actor. No outcome-based exclusions are made. The policies and clean tuning thresholds were frozen before external feature inference. All entries below are percentages or accuracy points.

Method	Acc.	Gain	Change	R	H	Neutral
af_gain	32.47	-1.91	20.83	3.99	5.90	10.94
scores_gain	33.16	-1.22	14.41	2.60	3.82	7.99
reliable_gain	32.64	-1.74	8.68	0.87	2.60	5.21
outcome_raw	32.12	-2.26	9.55	0.69	2.95	5.90
outcome_isotonic	34.55	0.17	1.04	0.17	0.00	0.87
two_raw	32.81	-1.56	11.63	1.74	3.30	6.60
two_isotonic	33.33	-1.04	8.51	1.22	2.26	5.03
two_platt	33.68	-0.69	6.25	0.52	1.22	4.51
supported_gain	34.38	0.00	0.00	0.00	0.00	0.00
certified_gain	34.38	0.00	0.00	0.00	0.00	0.00
margin	30.38	-3.99	34.55	6.08	10.07	18.40
confidence	29.69	-4.69	36.81	6.42	11.11	19.27
entropy	27.43	-6.94	41.84	6.60	13.54	21.70
legacy_opportunity_advantage	26.74	-7.64	52.95	8.51	16.15	28.30
legacy_logistic	27.60	-6.77	46.88	7.64	14.41	24.83
legacy_gb_gain	29.34	-5.03	40.28	6.77	11.81	21.70
legacy_rf_gain	31.77	-2.60	28.47	5.21	7.81	15.45
legacy_mlp	29.17	-5.21	53.30	9.38	14.58	29.34

Positive at-most batch, same nominal budget

Method	Gain	CI low	CI high	Change
af_gain	-1.39	-3.99	0.52	19.97
scores_gain	-1.22	-3.30	0.52	14.41
reliable_gain	-1.74	-3.30	-0.35	8.68
outcome_raw	-2.26	-3.82	-0.87	9.55
outcome_isotonic	0.17	0.00	0.52	1.04
two_raw	-1.56	-3.65	0.17	11.63
two_isotonic	-1.04	-2.78	0.52	8.51
two_platt	-0.69	-1.74	0.35	6.25
supported_gain	0.00	0.00	0.00	0.00
certified_gain	0.00	0.00	0.00	0.00
margin	-0.87	-3.12	1.39	19.97
confidence	-1.56	-3.47	0.35	19.97
entropy	-3.12	-5.38	-0.87	19.97

Fixed-threshold intervals resample paired actors with the policy fixed. Batch intervals recompute the rank-constrained selection within each resampled batch. They are conditional on frozen experts and router fitting. The two modes do not have identical deployment semantics. Full budget sweeps and raw endpoints are in the CSV files.

S12. External RAVDESS: seed 11, video_moving_occluder

The subset contains 576 clips from 24 actors, with 24 clips per actor. No outcome-based exclusions are made. The policies and clean tuning thresholds were frozen before external feature inference. All entries below are percentages or accuracy points.

Method	Acc.	Gain	Change	R	H	Neutral
af_gain	33.16	-1.22	12.15	1.74	2.95	7.47
scores_gain	33.16	-1.22	12.85	1.91	3.12	7.81
reliable_gain	34.03	-0.35	2.78	0.35	0.69	1.74
outcome_raw	32.99	-1.39	9.38	1.04	2.43	5.90
outcome_isotonic	34.20	-0.17	0.35	0.00	0.17	0.17
two_raw	34.03	-0.35	11.28	1.91	2.26	7.12
two_isotonic	33.68	-0.69	7.99	1.04	1.74	5.21
two_platt	33.68	-0.69	6.42	0.69	1.39	4.34
supported_gain	34.38	0.00	0.00	0.00	0.00	0.00
certified_gain	34.38	0.00	0.00	0.00	0.00	0.00
margin	32.81	-1.56	22.40	4.51	6.08	11.81
confidence	32.47	-1.91	24.48	4.51	6.42	13.54
entropy	30.21	-4.17	26.22	4.34	8.51	13.37
legacy_opportunity_advantage	28.99	-5.38	45.49	7.12	12.50	25.87
legacy_logistic	30.90	-3.47	35.59	6.42	9.90	19.27
legacy_gb_gain	31.60	-2.78	29.51	4.34	7.12	18.06
legacy_rf_gain	33.68	-0.69	8.68	1.04	1.74	5.90
legacy_mlp	30.73	-3.65	42.01	7.81	11.46	22.74

Positive at-most batch, same nominal budget

Method	Gain	CI low	CI high	Change
af_gain	-1.22	-2.78	0.35	12.15
scores_gain	-1.22	-2.95	0.17	12.85
reliable_gain	-0.35	-1.04	0.35	2.78
outcome_raw	-1.39	-2.78	-0.17	9.38
outcome_isotonic	-0.17	-0.52	0.00	0.35
two_raw	-0.35	-1.91	0.87	11.28
two_isotonic	-0.69	-2.08	0.35	7.99
two_platt	-0.69	-2.08	0.35	6.42
supported_gain	0.00	0.00	0.00	0.00
certified_gain	0.00	0.00	0.00	0.00
margin	-1.91	-4.51	0.87	19.97
confidence	-1.74	-4.51	0.52	19.97
entropy	-3.30	-5.90	-1.22	19.97

Fixed-threshold intervals resample paired actors with the policy fixed. Batch intervals recompute the rank-constrained selection within each resampled batch. They are conditional on frozen experts and router fitting. The two modes do not have identical deployment semantics. Full budget sweeps and raw endpoints are in the CSV files.

S13. External RAVDESS: seed 22, clean

The subset contains 576 clips from 24 actors, with 24 clips per actor. No outcome-based exclusions are made. The policies and clean tuning thresholds were frozen before external feature inference. All entries below are percentages or accuracy points.

Method	Acc.	Gain	Change	R	H	Neutral
af_gain	40.80	5.21	9.38	5.90	0.69	2.78
scores_gain	41.49	5.90	11.98	7.12	1.22	3.65
reliable_gain	41.15	5.56	10.24	6.25	0.69	3.30
outcome_raw	41.49	5.90	11.81	7.12	1.22	3.47
outcome_isotonic	39.41	3.82	5.56	3.99	0.17	1.39
two_raw	41.32	5.73	15.45	7.81	2.08	5.56
two_isotonic	39.93	4.34	11.28	5.73	1.39	4.17
two_platt	41.15	5.56	14.93	7.64	2.08	5.21
supported_gain	35.59	0.00	0.00	0.00	0.00	0.00
certified_gain	35.59	0.00	0.00	0.00	0.00	0.00
margin	41.49	5.90	16.67	8.33	2.43	5.90
confidence	41.49	5.90	17.53	8.33	2.43	6.77
entropy	41.84	6.25	17.71	8.85	2.60	6.25

Positive at-most batch, same nominal budget

Method	Gain	CI low	CI high	Change
af_gain	5.21	3.12	7.12	9.38
scores_gain	5.90	3.30	8.51	11.98
reliable_gain	5.56	3.30	7.99	10.24
outcome_raw	5.90	3.65	8.33	11.81
outcome_isotonic	3.82	1.91	5.90	5.56
two_raw	5.73	3.30	8.33	15.45
two_isotonic	4.34	2.26	6.60	11.28
two_platt	5.56	3.12	8.16	14.93
supported_gain	0.00	0.00	0.00	0.00
certified_gain	0.00	0.00	0.00	0.00
margin	5.90	2.60	9.03	16.67
confidence	5.90	2.78	9.03	17.53
entropy	6.25	2.78	9.20	17.71

Fixed-threshold intervals resample paired actors with the policy fixed. Batch intervals recompute the rank-constrained selection within each resampled batch. They are conditional on frozen experts and router fitting. The two modes do not have identical deployment semantics. Full budget sweeps and raw endpoints are in the CSV files.

S14. External RAVDESS: seed 22, video_occlude50

The subset contains 576 clips from 24 actors, with 24 clips per actor. No outcome-based exclusions are made. The policies and clean tuning thresholds were frozen before external feature inference. All entries below are percentages or accuracy points.

Method	Acc.	Gain	Change	R	H	Neutral
af_gain	33.68	-1.91	23.44	4.69	6.60	12.15
scores_gain	30.21	-5.38	50.17	11.46	16.84	21.88
reliable_gain	34.03	-1.56	23.61	6.60	8.16	8.85
outcome_raw	34.38	-1.22	19.79	5.38	6.60	7.81
outcome_isotonic	36.63	1.04	10.76	3.65	2.60	4.51
two_raw	35.42	-0.17	17.01	5.38	5.56	6.08
two_isotonic	35.42	-0.17	12.50	3.82	3.99	4.69
two_platt	35.07	-0.52	16.32	5.03	5.56	5.73
supported_gain	35.59	0.00	0.00	0.00	0.00	0.00
certified_gain	35.59	0.00	0.00	0.00	0.00	0.00
margin	29.51	-6.08	58.16	11.81	17.88	28.47
confidence	30.21	-5.38	55.90	11.46	16.84	27.60
entropy	29.51	-6.08	55.90	11.28	17.36	27.26

Positive at-most batch, same nominal budget

Method	Gain	CI low	CI high	Change
af_gain	-1.56	-3.65	0.53	19.97
scores_gain	-1.39	-3.65	1.22	19.97
reliable_gain	-0.17	-2.60	2.60	19.97
outcome_raw	-1.22	-3.65	1.39	19.79
outcome_isotonic	1.04	-0.69	2.78	10.76
two_raw	-0.17	-2.26	1.74	17.01
two_isotonic	-0.17	-2.08	1.56	12.50
two_platt	-0.52	-2.43	1.39	16.32
supported_gain	0.00	0.00	0.00	0.00
certified_gain	0.00	0.00	0.00	0.00
margin	-0.35	-2.78	2.43	19.97
confidence	-0.87	-3.30	2.08	19.97
entropy	-0.87	-3.99	1.74	19.97

Fixed-threshold intervals resample paired actors with the policy fixed. Batch intervals recompute the rank-constrained selection within each resampled batch. They are conditional on frozen experts and router fitting. The two modes do not have identical deployment semantics. Full budget sweeps and raw endpoints are in the CSV files.

S15. External RAVDESS: seed 22, video_moving_occluder

The subset contains 576 clips from 24 actors, with 24 clips per actor. No outcome-based exclusions are made. The policies and clean tuning thresholds were frozen before external feature inference. All entries below are percentages or accuracy points.

Method	Acc.	Gain	Change	R	H	Neutral
af_gain	34.03	-1.56	14.93	2.78	4.34	7.81
scores_gain	33.33	-2.26	34.55	9.03	11.28	14.24
reliable_gain	33.16	-2.43	36.46	9.90	12.33	14.24
outcome_raw	32.81	-2.78	30.56	7.81	10.59	12.15
outcome_isotonic	34.90	-0.69	18.23	4.51	5.21	8.51
two_raw	31.25	-4.34	41.67	10.24	14.58	16.84
two_isotonic	31.94	-3.65	35.76	8.68	12.33	14.76
two_platt	31.08	-4.51	41.32	10.07	14.58	16.67
supported_gain	35.59	0.00	0.00	0.00	0.00	0.00
certified_gain	35.59	0.00	0.00	0.00	0.00	0.00
margin	33.85	-1.74	17.53	4.34	6.08	7.12
confidence	33.51	-2.08	17.88	4.34	6.42	7.12
entropy	33.85	-1.74	18.40	4.34	6.08	7.99

Positive at-most batch, same nominal budget

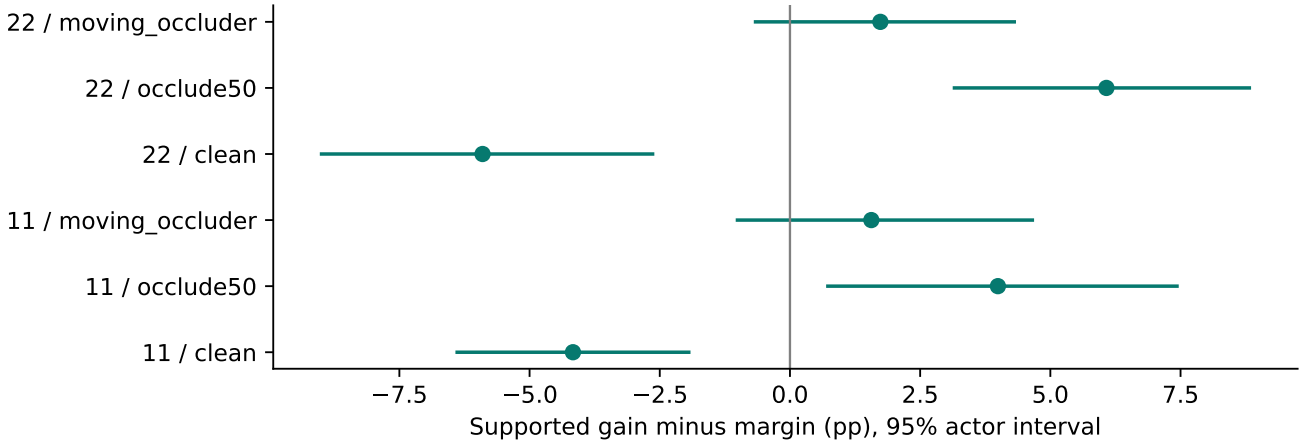
Method	Gain	CI low	CI high	Change
af_gain	-1.56	-4.51	1.04	14.93
scores_gain	-0.52	-3.65	2.43	19.97
reliable_gain	-2.43	-5.03	1.22	19.97
outcome_raw	-1.39	-4.86	2.26	19.97
outcome_isotonic	-0.69	-3.65	2.60	18.23
two_raw	-2.08	-4.86	0.87	19.97
two_isotonic	-2.60	-5.03	0.35	19.97
two_platt	-2.78	-5.03	0.35	19.97
supported_gain	0.00	0.00	0.00	0.00
certified_gain	0.00	0.00	0.00	0.00
margin	-1.74	-4.34	0.69	17.53
confidence	-2.08	-4.34	0.17	17.88
entropy	-1.74	-4.86	0.87	18.40

Fixed-threshold intervals resample paired actors with the policy fixed. Batch intervals recompute the rank-constrained selection within each resampled batch. They are conditional on frozen experts and router fitting. The two modes do not have identical deployment semantics. Full budget sweeps and raw endpoints are in the CSV files.

S16. Primary and secondary paired contrasts

Support-aware minus comparator, at the nominal 20%fixed-threshold operating point. Only clean seed 11 versus margin is the locked primary comparison. Other entries are secondary diagnostics, not substitute endpoints after observing failure. Intervals are paired actor bootstrap estimates, not exact finite-sample guarantees.

Seed / condition / comparator	Diff.	CI low	CI high
11/clean/margin	-4.17	-6.42	-1.91
11/clean/reliable_gain	-0.35	-0.87	0.00
11/clean/outcome_raw	-0.87	-2.26	0.17
11/occlude50/margin	3.99	0.69	7.47
11/occlude50/reliable_gain	1.74	0.35	3.30
11/occlude50/outcome_raw	2.26	0.87	3.82
11/moving_occluder/margin	1.56	-1.04	4.69
11/moving_occluder/reliable_gain	0.35	-0.35	1.04
11/moving_occluder/outcome_raw	1.39	0.17	2.78
22/clean/margin	-5.90	-9.03	-2.60
22/clean/reliable_gain	-5.56	-7.99	-3.30
22/clean/outcome_raw	-5.90	-8.33	-3.65
22/occlude50/margin	6.08	3.12	8.85
22/occlude50/reliable_gain	1.56	-0.87	3.99
22/occlude50/outcome_raw	1.22	-1.39	3.99
22/moving_occluder/margin	1.74	-0.69	4.34
22/moving_occluder/reliable_gain	2.43	-1.39	6.08
22/moving_occluder/outcome_raw	2.78	-0.87	6.77



S17. Every external actor at the primary operating point

Each row is an independent-source unit for the reported bootstrap, under the working assumption that actors are independent. Clips within an actor do not increase the number of independent sources. Values are gain points over audio. The same actor remains in the analysis in every condition and for every policy.

Actor	Clean S	Clean M	Gray S	Gray M	Moving S	Moving M
100.00	0.00	12.50	0.00	-25.00	0.00	-25.00
200.00	0.00	0.00	0.00	12.50	0.00	8.33
300.00	0.00	0.00	0.00	-4.17	0.00	0.00
400.00	0.00	0.00	0.00	-4.17	0.00	-4.17
500.00	0.00	8.33	0.00	-4.17	0.00	-4.17
600.00	0.00	4.17	0.00	-8.33	0.00	0.00
700.00	0.00	16.67	0.00	4.17	0.00	4.17
800.00	0.00	8.33	0.00	-8.33	0.00	-4.17
900.00	0.00	8.33	0.00	-16.67	0.00	-8.33
1000.00	0.00	0.00	0.00	-12.50	0.00	0.00
1100.00	0.00	0.00	0.00	8.33	0.00	0.00
1200.00	0.00	-4.17	0.00	-4.17	0.00	4.17
1300.00	0.00	8.33	0.00	4.17	0.00	4.17
1400.00	0.00	-4.17	0.00	-12.50	0.00	-8.33
1500.00	0.00	-4.17	0.00	-8.33	0.00	-4.17
1600.00	0.00	8.33	0.00	4.17	0.00	4.17
1700.00	0.00	0.00	0.00	-12.50	0.00	-4.17
1800.00	0.00	8.33	0.00	-4.17	0.00	4.17
1900.00	0.00	8.33	0.00	0.00	0.00	8.33
2000.00	0.00	8.33	0.00	0.00	0.00	-4.17
2100.00	0.00	8.33	0.00	0.00	0.00	0.00
2200.00	0.00	0.00	0.00	0.00	0.00	-12.50
2300.00	0.00	-4.17	0.00	-8.33	0.00	-4.17
2400.00	0.00	8.33	0.00	4.17	0.00	8.33

S denotes support-aware direct gain; M denotes the margin gate. Worst-source values are retained, not filtered. The second checkpoint is supplied in source_results.csv. These data describe the selected actors and task; they do not establish a demographic or natural-population guarantee.

S18. Calibration in the switching region

Calibration is evaluated both on all disagreements and on the positive-score switching region. Regions are outcome-free score selections, not chosen for favorable empirical calibration. The table aggregates occupied equal-width probability bins on seed 11; weighted absolute calibration error is descriptive and depends on binning. It is not a bound under shift.

Condition / map / outcome / region	N	Pred.	Observed	Bin error
clean/direct/harm/all	218	50.89	17.43	33.46
clean/direct/harm/positive	2	23.92	0.00	23.92
clean/direct/rescue/all	218	15.25	39.45	24.99
clean/direct/rescue/positive	2	43.25	0.00	43.25
clean/iso/harm/all	218	51.46	17.43	34.02
clean/iso/harm/positive	22	29.10	4.55	24.55
clean/iso/rescue/all	218	16.59	39.45	24.12
clean/iso/rescue/positive	22	34.21	27.27	6.94
clean/platt/harm/all	218	52.27	17.43	34.84
clean/platt/harm/positive	18	27.97	0.00	27.97
clean/platt/rescue/all	218	17.35	39.45	24.20
clean/platt/rescue/positive	18	36.54	27.78	13.47
moving_occluder/direct/harm/all	434	51.68	32.03	19.65
moving_occluder/direct/harm/positive	2	23.17	50.00	26.83
moving_occluder/direct/rescue/all	434	15.59	16.59	4.61
moving_occluder/direct/rescue/positive	2	41.89	0.00	41.89
moving_occluder/iso/harm/all	434	50.42	32.03	18.47
moving_occluder/iso/harm/positive	46	28.30	21.74	9.14
moving_occluder/iso/rescue/all	434	16.30	16.59	7.80
moving_occluder/iso/rescue/positive	46	35.45	13.04	24.85
moving_occluder/platt/harm/all	434	52.39	32.03	21.24
moving_occluder/platt/harm/positive	37	29.40	21.62	12.66
moving_occluder/platt/rescue/all	434	16.92	16.59	7.08
moving_occluder/platt/rescue/positive	37	36.03	10.81	25.22
occlude50/direct/harm/all	451	59.14	32.82	26.66
occlude50/direct/harm/positive	6	24.92	0.00	24.92
occlude50/direct/rescue/all	451	12.77	15.52	7.67
occlude50/direct/rescue/positive	6	43.03	16.67	26.37
occlude50/iso/harm/all	451	57.61	32.82	25.00
occlude50/iso/harm/positive	49	28.66	26.53	7.32
occlude50/iso/rescue/all	451	15.15	15.52	6.73
occlude50/iso/rescue/positive	49	36.76	14.29	22.48
occlude50/platt/harm/all	451	55.58	32.82	22.76
occlude50/platt/harm/positive	36	27.98	19.44	8.53
occlude50/platt/rescue/all	451	16.58	15.52	6.19
occlude50/platt/rescue/positive	36	38.63	8.33	30.30

Counts are clips, not independent sources. High-score selection is specific to the corresponding calibrated score; it is not necessarily the support-aware policy's final fixed-threshold mask. Final decision quality is reported separately as selected gain, harm and volume in confirmation_results.csv.

S19. Bootstrap implementation, power, and inference cost

For fixed thresholds, compute one mean outcome per actor and average the paired actor resamples. For a batch rule, repeated actors create repeated copies of their clips. Sort the repeated copies by frozen score and deterministic clip identity, keep positive scores only, and reapply the budget in each bootstrap batch. Equal 24-clip actor groups permit an equivalent multiplicity implementation: cumulative copies in score order are truncated at the batch cap. Tests compare that implementation with explicit repeated-clip ranking.

The draw matrix has 5,000 rows and 24 actor draws per row, generated with seed 56013. All methods share the draw matrix. Paired comparator intervals use the bootstrap difference, not subtraction of unrelated interval endpoints. The method's training randomness is conditioned on, and checkpoint 11/22 comparisons are reported separately rather than treated as independent 24-source replications.

A planning approximation uses the inspected CREMA candidate-minus-margin actor standard deviation 0.04148. With 24 sources, two-sided 5%alpha and 80%power, $(1.96 + 0.842)\sigma/\sqrt{24}$ is about 2.37 points. A 2-point effect would require about 34 sources under the same approximation. The external domain can change variance, so this is planning information rather than a guaranteed-power assertion.

Encoder and head measurements are separate from selector overhead. Raw extraction records audio-encoder time and video-encoder time including the four crop probes. The original video path uses eight frames; the reliability path encodes twelve images. Shared forward batches prevent the logged combined visual time from being interpreted as a separately measured incremental cost. Head timing executes all three heads on 576 cached feature vectors. Selector timing executes the new comparator suite as a batch. No number is presented as streaming latency or an equal-total-compute frontier.

Component	Median seconds	Mean seconds	Calls
audio_encoder	0.0521	0.0716	576
visual_encoder_with_reliability_probes	0.3833	0.3938	1728

S20. Reproduction, demonstrations, and unresolved requirements

Portable route replay. Install the recorded Python requirements, then run test_policies.py and replay.py. The replay loads trusted locally produced fitted models, recomputes scores from bundled logits and reliability measurements, checks all released masks, and verifies gains against the result tables. It needs no raw video or GPU. Calibration and region-support audit inputs are included. The original revision 55 models and their locks remain unchanged.

Full extraction. The source manifests identify official media, selected filenames, member CRC and local SHA values. Download scripts support resuming incomplete transfers. Raw feature extraction requires the pinned WavLM and DINOv2 weights and sources. Expert checkpoints and normalization parameters are copied into the package, with provenance. Execution scripts retain the original workspace layout; path adaptation is documented. No pretrained encoder weights or complete raw dataset are included in the ZIP.

Demonstrations. The static site includes real source videos, measured scores and decisions, condition controls, shared playback, and links to the complete tables. Illustrative case selection is explicit and deterministic. Neither the score curve nor playback claims streaming model updates. Derived RAVDESS media retain the dataset's noncommercial share-alike license and attribution.

Review response. The positive-gain decision correction, calibration plateau audit, less flexible calibration, direct and three-outcome controls, raw-video reliability signals, two checkpoint evaluations, fresh external source reserve, batch re-ranking bootstrap, paired contrasts, power calculation, and real-video demonstrations are implemented. The old locked superiority result is not rewritten retrospectively.

Still missing. There is no newly collected CREMA confirmation population, no natural-occlusion dataset, no physical Ameca test, and no guarantee of top-tier novelty. The formal region bound certifies no switches at this source count. A support detector that simply retains audio under severe corruption must be judged by its clean opportunity cost and independent performance, not by zero switch-induced harm alone. The local primary result fails: the support gate rejects all external videos, including clean inputs, and trails margin by 4.17 points (95% actor interval [-6.42, -1.91]). This is not a successful transferable remedy. The separate scene-native workbench replays three TCOW videos with physical occluders, archived learned trackers and a causal template control; it does not validate the emotion selector. The edited speaker-pair follow-up is exploratory and uses extra identity references. Any next model chosen using this now-inspected external test needs a new independent reserve.